Proficient in NX CAD (80 Hours)

S.No	Session	Topics	Type	Duration (Hours)
1	Session 1	**INTRODUCTION TO NX-CAD** **Basic Terminologies:** CAD, CAM, CAE, PDM and Design Process Introduction to Unigraphics NX , About NX Gateway, Getting Started **NX Graphical User Interface:** Title bar, Menu bar, Toolbar, Radial toolbar Selection bar, Cue and status line Resource bar, Navigators, HD3D tools Integrated browser, Palettes, Roles, Full screen **View** **orientation-** trimetric, isometric, View commands Rotate ,Pan, Zoom in/out, Quick pick, Quick pick, categories **Coordinate system** absolute coordinate system, WCS, Absolute coordinate, Work coordinate system. View triad, Multiple graphics window, Information window Keyboard accelerators, Dialog box **File management:** Creating new files and about templates, Opening files, Saving files	Theory/Demo	2.0Hrs
2	Session 2	Exercises included for Practice	Practicals	2.0Hrs
3	Session 3	**SKETCH CREATION & EDITING TOOLS** **Creating Sketches:** Profile, Line, Arc, Circle, Fillet, Chamfer, Rectangle, Polygon, Studio Spline, Fit spline, Ellipse, Conic. **Editing sketches:** Quick trim, Quick extend, Make corner, Offset curve, Pattern curve, Mirror curve, Intersection point, Derived lines **Constraints:** Geometric constraints, Auto constraint, Inferred constraint Dimensional constraints, Auto dimension, Continuous auto dimension Animate dimension	Theory/Demo	2.0Hrs
4	Session 4	Exercises included for Practice	Practicals	2.0Hrs
5	Session 5	**PART MODELING FEATURES - I** **Basic terminologies:** Feature, Body, Solid body, Sheet, Face, Section curves, Guide curves Feature modeling concepts About Datum CSYS and Datum Planes, Changing units in NX **Creating Extrude features:** Extrusion by selecting a section of edges Limits option Boolean operation Applying draft and Offset sketches	Theory/Demo	2.0Hrs
6	Session 6	Exercises included for Practice	Practicals	2.0Hrs
7	Session 7	**Creating Revolve features:** Revolve sketch about an axis Limits option Offset sketches **Creating Primitives:** Block Cylinder Cone Sphere	Theory/Demo	2.0Hrs
8	Session 8	Exercises included for Practice	Practicals	2.0Hrs
9	Session 9	**Creating Datum Features:** Datum Plane Datum axis Datum CSYS Datum Point	Theory/Demo	2.0Hrs
10	Session 10	Exercises included for Practice	Practicals	2.0Hrs
11	Session 11	**Creating Sweep Features:** Sweep a section along a guide Adding thickness Sweep one or more sections along guide curves Section sweep using spine Section location Interpolation Alignment Orientation method Scaling method	Theory/Demo	2.0Hrs
12	Session 12	Exercises included for Practice	Practicals	2.0Hrs
		Creating Variational Sweep		

13	Session 13	Limits option Adding secondary sections **Creating Tube feature** **Blend Features:** Creating Edge blend Constant radius Variable radius Creating Face blend Types of blend	Theory/Demo	2.0Hrs
14	Session 14	Exercises included for Practice	Practicals	2.0Hrs
15	Session 15	**PART EDITING FEATURES - I** Feature Modeling Commands **Hole:** General hole Drill size holes Screw clearance holes Threaded holes **Boss** **Pocket:** Cylindrical, Rectangular, General **Pad:** Rectangular, General Applying Chamfer and Fillet, Dart, Thread, Shell **Instance feature:** Rectangular array, Circular array, Pattern face Mirror feature, Mirror body	Theory/Demo	2.0Hrs
16	Session 16	Exercises included for Practice	Practicals	2.0Hrs
17	Session 17	**PART EDITING FEATURES - II** **Emboss:** Offset emboss **Slot:** Rectangular, Ball end, U-Slot, T-Slot, Dove tail **Groove:** Rectangular, Ball end, U-groove **Draft:** From plane From edges Tangent to faces To parting edges Draw direction Variable draft points Draft body, Scale To Divide face, Trim body , Split body Boolean commands	Theory/Demo	2.0Hrs
18	Session 18	Exercises included for Practice	Practicals	2.0Hrs
19	Session 19	**User defined feature** Creating Feature group **Editing Expressions** Knowing Feature replay command Knowing Layer settings To measure distance between geometries To measure angle between geometries To measure bodies and face geometries To find geometric properties	Theory/Demo	2.0Hrs
20	Session 20	Exercises included for Practice	Practicals	2.0Hrs
21	Session 21	**ASSEMBLY MODELING - I** **Feature operations and smart tools:** Synchronous Modeling Move Face, Pull-Face, Offset Region, Resize Face Delete Face, Resize Blend, Reorder Blends, Replace Blend Label Chamfer, Resize Chamfer Cut Face, Copy Face, Paste Face Mirror Face, Pattern Face Move Edge, Offset Edge, Edit Cross Section Make Coplanar, Linear Dimension **Introduction to Assembly modeling** **Types of approach:** Bottom up assembly modeling, Top down assembly modeling **Placing components**	Theory/Demo	2.0Hrs
22	Session 22	Exercises included for Practice	Practicals	2.0Hrs
23	Session 23	**Assembly Constraints** Angle, Bond, Center, Concentric, Distance, Fit, Parallel, Perpendicular, Touch align Fix, Align/ Lock, Bond **Creating component arrays** Linear array, Circular array, Feature instance array	Theory/Demo	2.0Hrs
24	Session 24	Exercises included for Practice	Practicals	2.0Hrs
		Assembly Modeling Moving a component, Replacing component, Repositioning component		

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25	Session 25	Mirroring assembly Creating a New Component, Creating new parent, Finding assembly clearance Finding degrees of freedom Creating exploded views Assembly sequencing with motion Creating deformable parts Assembly envelopes	Theory/Demo	2.0Hrs
26	Session 26	Exercises included for Practice	Practicals	2.0Hrs
27	Session 27	**Introduction to drawing** **Inserting new sheets, Editing sheets** **Setting up standards, Knowing Graphical User Interface of drafting** **NX drafting methods** **Creating drafting views** Base view Drawing view Projected view Orthographic view Auxiliary view	Theory/Demo	2.0Hrs
28	Session 28	Exercises included for Practice	Practicals	2.0Hrs
29	Session 29	**Creating drafting views** **Detail view** **Section view** Simple section, Stepped section , Half section, Revolved section Folded section, Unfolded section Pictorial section, Half pictorial section Break out section Creating Broken view Cropping view boundary, Standard view **How to move/Copy a view** **How to align a view** **How to hide/show components** **Creating section in view**	Theory/Demo	2.0Hrs
30	Session 30	Exercises included for Practice	Practicals	2.0Hrs
31	Session 31	**Adding dimensions** Inferred Dimension Horizontal Dimension, Vertical Dimension Parallel Dimension, Perpendicular dimension, Angular dimension Cylindrical Dimension, Hole dimension, Diameter Dimension Chamfer Dimension, Radius or Radius of Curvature Dimension, Radius to Centre, Folded Radius, Thickness Dimension, Arc Length Horizontal Chain Dimension, Vertical Chain Dimension Horizontal Baseline Dimension, Vertical Baseline Dimension, Ordinate Dimension **Adding annotations** **Adding Feature control frame** **Adding Datum Feature symbol** **Adding Datum Target** **Inserting Identification symbol** **Inserting Surface Finish symbol** **Placing target point symbol** **Placing Intersection symbol** **Applying cross hatch and area fill** **Creating centrelines** **Placing table** **Placing tabular note** **Listing parts list** **Creating auto balloon**	Theory/Demo	2.0Hrs
32	Session 32	Exercises included for Practice	Practicals	2.0Hrs
33	Session 33	**Surface Modeling commands** Creating extrude surface Creating revolved surface Creating ruled surface **Surface using curves** Surface by Through curves Surface by Through curve mesh Creating Studio surface Surface from Section Surface Surface creation by N-Sided surface	Theory/Demo	2.0Hrs
34	Session 34	Exercises included for Practice	Practicals	2.0Hrs
		Surface generation		

		Creating surface using Styled Sweep, Surface from Four Point Surface		
		Swoop feature, Transition feature, Extension feature		
		Bounded plane, Sheet from curves, Ribbon builder,		
		Patch openings, Law extension, Silhouette flange		
		Creating curves from curves		
35	Session 35	**Creating curve from bodies**	Theory/Demo	2.0Hrs
		Creating associative copies		
		Extract body, Composite curve		
		Editing geometries		
		Emboss sheet, Sew and Unsew, Patch, Trim body, Split body		
		Trimmed sheet, Trim and extend, Untrim, Offset surface		
		Variable offset surface, Offset face, Scale body, Thicken, Divide face		
36	Session 36	Exercises included for Practice	Practicals	2.0Hrs
		Sheet metal Design		
		About NX Sheet Metal Preferences		
		Creating base feature		
		Tab		
		Creating Bend feature		
		Attaching flange, Attaching Contour flange, Creating Lofted flange		
		Inserting Hem flange		
		Apply Bend, unbend, rebend		
		Apply Jog, Creating Sheet metal from solid		
		Corner definition		
37	Session 37	Applying Closed corner, Break corner, Applying chamfer	Theory/Demo	2.0Hrs
		Applying punch operations		
		Dimple, Louver, drawn cutout, bead, Solid punch, gusset		
		Sheet metal cut operations		
		Resizing bend radius		
		Resizing bend angles		
		Resizing Neutral Factor		
		Bend operations		
		Converting solid body to sheet metal		
		Converting to sheet metal		
		Creating flat pattern		
38	Session 38	Exercises included for Practice	Practicals	2.0Hrs
		RENDERING		
		INTRODUCTION:		
		Rendering Techniques, Rendering in NX CAD, Rendering Workflow overview		
		Rendering Modes:		
39	Session 39	Advanced Studio, Ray Traced Studio	Theory/Demo	2.0Hrs
		True shading setup:		
		Global Reflection, Object Materials, Backgrounds, Show Shadows		
		Show Floor Reflection, Show Floor Grid, Face Edges, Scene Lighting		
		System Studio Enhancements:		
		System Studio Materials, System Scenes, Export		
40	Session 40	Exercises included for Practice	Practicals	2.0Hrs